Jaypee Institute of Information Technology, Noida
Quantum Mechanics for Engineers (26M12PH111) [2026]
Tute-2

1. If $\hat{x}$ and $\hat{p}_x$ are position and momentum operators, prove that $[\hat{x}, {\hat{p}_x}^n] = ni\hbar {\hat{p}_x}^{n-1}$. **[CO2]**

2. If $\hat{x}$ and $\hat{p}_x$ are position and momentum operators, prove that $[\hat{x}^n, \hat{p}_x] = ni\hbar \hat{x}^{n-1}$. **[CO2]**

3. Solve the commutation relation $\left[\hat{x}, \hat{H}\right]$ and $\left[\hat{x}, \left[\hat{x}, \hat{H}\right]\right]$. **[CO2]**

4. Using the usual canonical commutation relations derive [A,B] where A= i(xp_y-yp_x) and B= i(yp_z-zp_y). **[CO2]**

5. If $\psi(x) = Ae^{-x^4}$ is the eigen function of a one dimensional Hamiltonian with eigen value E= 0, determine the value of potential V(x) in the units where $\hbar = 2m = 1$. **[CO3]**

6. Operator $a = x + \frac{d}{dx}$ derive [a,cosx]. **[CO2]**

7. Solve [x^2, p^2]. **[CO2]**

8. The momentum operator is defined as $\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$, determine which of the following functions are eigenfunction of momentum operator: sinkx, e^{ikx}, x^4. **[CO2]**

9. The state of a one dimensional quantum system is represented by a wave function $\psi(x) = N \sin(3\pi x)$ for 0<x<1 with N being the normalization factor. Calculate the probability that a measurement of the particle will give a result in the range 2/3<x<1. **[CO3].**